

Базы данных: От поиска до автоматизации

Практическое руководство по работе с NCBI, BLAST и Biopython.





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Resource List (A-Z)

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Welcome to NCBI

The National Center for Biotechnology Information advances science and health by providing access to biomedical and genomic information.

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Find help documents, attend a
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Develop

Use NCBI APIs and code
libraries to build applications



Analyze

Identify an NCBI tool for your
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Explore NCBI research and
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Popular Resources

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NCBI News & Blog

Now Available: RefSeq Release 233

19 Feb 2026

RefSeq release 233 is now
available online and from the FTP site!
You can access RefSeq data

Changes to PMC Article Dataset
Distribution Services Coming in 2026

12 Feb 2026

PMC will make major changes

NCBI — это не Google, а реляционная сеть

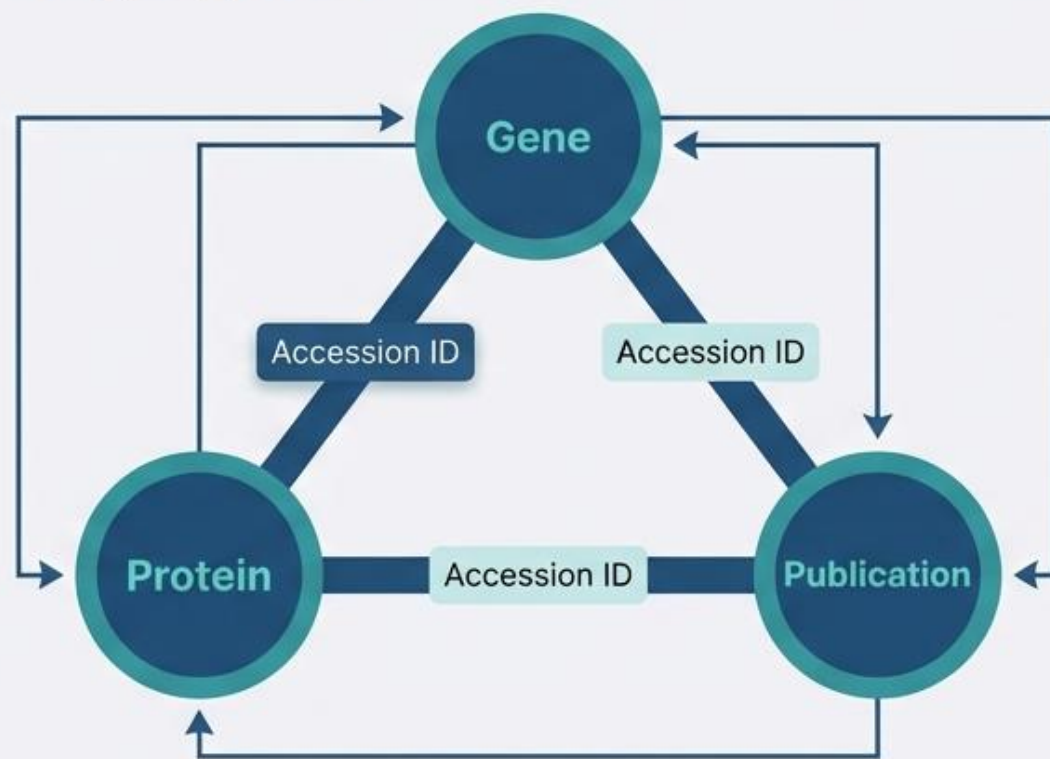
Обычный поиск по ключевым словам не работает эффективно, когда данных слишком много. NCBI – это не файловый архив, а сеть, связанная уникальными идентификаторами (Accession numbers).

Неверный подход



Недостаточно эффективно

Подход NCBI





Search

[Advanced](#)

[User Guide](#)

About PubMed

PubMed is a free resource supporting the search and retrieval of biomedical and life sciences literature with the aim of improving health—both globally and personally.

PubMed contains more than 39 million citations and abstracts of biomedical literature. It does not include full text journal articles; however, links to the full text are often present when available from other sources, such as the publisher's website or [PubMed Central \(PMC\)](#).

Available to the public online since 1996, PubMed was developed and is maintained by the [National Center for Biotechnology Information \(NCBI\)](#), at the [U.S. National Library of Medicine \(NLM\)](#), located at the [National Institutes of Health \(NIH\)](#).

About the Content

Citations in PubMed primarily stem from the biomedicine and health fields, and related disciplines such as life sciences, behavioral sciences, chemical sciences, and bioengineering.

PubMed facilitates searching across several NLM literature resources:

<https://pubmed.ncbi.nlm.nih.gov/>

Инструменты точного поиска

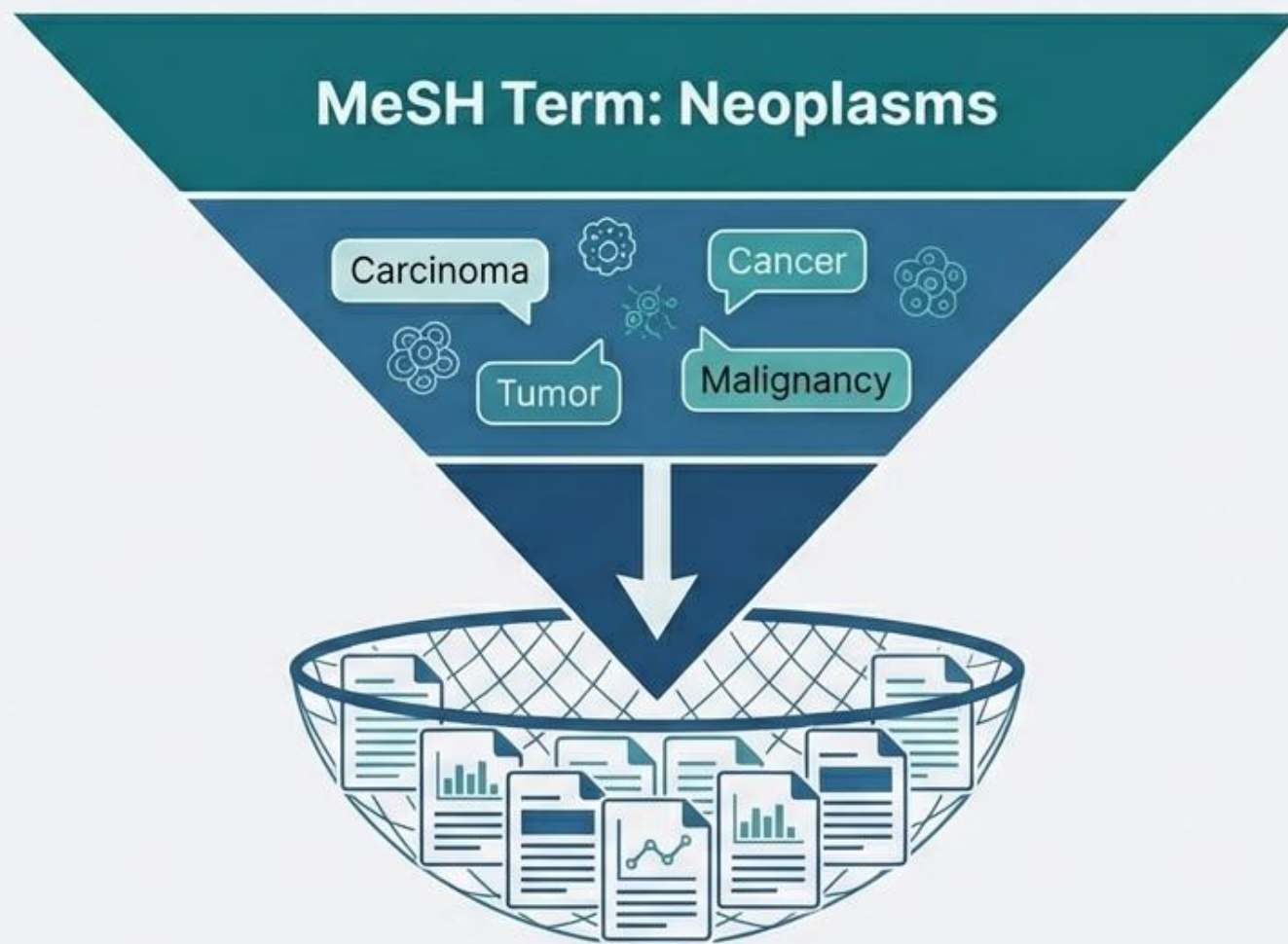
Используйте Advanced Search Builder для комбинирования полей. Логические операторы (AND, OR, NOT) позволяют отсеять лишнее.

The diagram illustrates the Advanced Search Builder interface. It consists of three main horizontal sections within a white rounded rectangle. The top section is a dark blue bar containing a teal dropdown menu labeled 'Title/Abstract' with a white downward arrow, and a dark blue text input field containing the text 'CRISPR'. An orange arrow points from the top left towards the 'Title/Abstract' dropdown. The middle section is a white bar containing an orange button labeled 'AND' with a white downward arrow. A long orange arrow points from the right towards this 'AND' button. The bottom section is another dark blue bar, similar to the top one, containing a teal dropdown menu labeled 'Journal' with a white downward arrow, and a dark blue text input field containing the text 'Nature'.

PubMed: Ищем смыслы, а не слова

Поиск по слову 'Cancer' упустит статьи с терминами 'Tumor' или 'Malignancy'.

Решение — MeSH (Medical Subject Headings). Алгоритм понимает иерархию понятий.





MeSH

MeSH ▾

Search

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MeSH

MeSH (Medical Subject Headings) is the NLM controlled vocabulary thesaurus used for indexing articles for PubMed.

Using MeSH

[Help](#)

[Tutorials](#)

More Resources

[E-Utilities](#)

[NLM MeSH Homepage](#)

<https://www.ncbi.nlm.nih.gov/mesh/>

GenBank против RefSeq: Хаос и Порядок

GenBank	RefSeq
	
Архив. Первичные данные, много дублей и ошибок. Все варианты сиквенсов.	Стандарт. Курируемые данные, без дублей. Эталонная последовательность.

Ищите префиксы **NM_** (мРНК) и **NP_** (белок) для проверенных данных.

Sample GenBank Record

This page presents an annotated sample GenBank record (accession number **U49845**) in its *GenBank Flat File* format. You can see the corresponding [live record for U49845](#), and see [examples of other records](#) that show a range of biological features.

GenBank Resources

[GenBank Home](#)

[Submission Types](#)

[Submission Tools](#)

[Search GenBank](#)

[Update GenBank Records](#)

LOCUS	SCU49845	5028 bp	DNA	PLN	21-JUN-1999
DEFINITION	Saccharomyces cerevisiae TCP1-beta gene, partial cds, and Axl2p (AXL2) and Rev7p (REV7) genes, complete cds.				
ACCESSION	U49845				
VERSION	U49845.1 GI:1293613				
KEYWORDS	.				
SOURCE	Saccharomyces cerevisiae (baker's yeast)				
ORGANISM	Saccharomyces cerevisiae Eukaryota; Fungi; Ascomycota; Saccharomycotina; Saccharomycetes; Saccharomycetales; Saccharomycetaceae; Saccharomyces.				
REFERENCE	1 (bases 1 to 5028)				
AUTHORS	Torpey,L.E., Gibbs,P.E., Nelson,J. and Lawrence,C.W.				
TITLE	Cloning and sequence of REV7, a gene whose function is required for DNA damage-induced mutagenesis in Saccharomyces cerevisiae				
JOURNAL	Yeast 10 (11), 1503-1509 (1994)				
PUBMED	7871890				
REFERENCE	2 (bases 1 to 5028)				
AUTHORS	Roemer,T., Madden,K., Chang,J. and Snyder,M.				
TITLE	Selection of axial growth sites in yeast requires Axl2p, a novel plasma membrane glycoprotein				
JOURNAL	Genes Dev. 10 (7), 777-793 (1996)				
PUBMED	8846915				
REFERENCE	3 (bases 1 to 5028)				
AUTHORS	Roemer,T.				
TITLE	Direct Submission				
JOURNAL	Submitted (22-FEB-1996) Terry Roemer, Biology, Yale University, New Haven, CT, USA				
FEATURES	Location/Qualifiers				
source	1..5028 /organism="Saccharomyces cerevisiae" /db_xref="taxon:4932" /chromosome="IX" /map="9"				
CDS	<1..206				

Homo sapiens insulin (INS), transcript variant 3, mRNA

NCBI Reference Sequence: [NM_001185098.2](#)

[FASTA](#) [Graphics](#)

Go to:

LOCUS	NM_001185098	644 bp	mRNA	linear	PRI 04-FEB-2026
DEFINITION	Homo sapiens insulin (INS), transcript variant 3, mRNA.				
ACCESSION	NM_001185098				
VERSION	NM_001185098.2				
KEYWORDS	RefSeq.				
SOURCE	Homo sapiens (human)				
ORGANISM	Homo sapiens Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo.				
REFERENCE	1 (bases 1 to 644)				
AUTHORS	Jabeen,A., Riaz,S., Usman,M., Parveen,A., Mukhtar,M., Wajid,A., Hanif,A. and Batool,A.				
TITLE	Association of polymorphism of NLRP3, ICAM-1, PTPN22, INS genes in childhood onset type 1 diabetes in a Pakistani population				
JOURNAL	Mol Biol Rep 51 (1), 1070 (2024)				
PUBMED	39425874				
REMARK	GeneRIF: Association of polymorphism of NLRP3, ICAM-1, PTPN22, INS genes in childhood onset type 1 diabetes in a Pakistani population. Publication Status: Online-Only				
REFERENCE	2 (bases 1 to 644)				
AUTHORS	Mao,C., Li,S., Che,J., Liu,D., Mao,X. and Rao,H.				
TITLE	The ubiquitin ligase UBR4 and the deubiquitylase USP5 modulate the stability of DNA mismatch repair protein MLH1				
JOURNAL	J Biol Chem 300 (8), 107592 (2024)				
PUBMED	39032648				
REFERENCE	3 (bases 1 to 644)				

- NM_** — (Mature mRNA)
Созревшая мРНК.
- NP_** — (Protein) Белковая последовательность.
- NR_** — (Non-coding RNA)
Некодирующая РНК (например, рРНК или тРНК).
- NC_** — (Chromosome) Целая хромосома.

Analyze this sequence

- Run BLAST
- Pick Primers
- Highlight Sequence Features
- Find in this Sequence
- Show in Genome Data Viewer

Reference sequence information

RefSeq alternative splicing
See 4 reference mRNA sequence splice variants for the INS gene.

More about the INS gene

This gene encodes insulin, a peptide hormone that plays a vital role in the regulation of carbohydrate and lipid metabolism. After removal o...
Also Known As: IDDM, IDDM1, IDDM2, ILP...

Related information

Protein

Eukaryotic genomes annotated at NCBI

Hundreds of eukaryotic genomes have been annotated by the NCBI Eukaryotic Genome Annotation Pipeline (see [graphs](#)). The latest annotation release available for each genome is shown in the tables below. The tables are organized by taxonomic group and provide links to the annotation report, FTP site, genome BLAST page, and Genome Data Viewer page.

Only completed annotations are shown here. Please browse the [annotation runs currently in progress](#) to see what will become available in a few days.

Show/Hide All

Featured (6)

FTP - FTP Download B - Organism-specific BLAST AR - Annotation Report GDV - Genome Data Viewer

Species	RefSeq assembly(ies)	Annotation Release	Freeze Date	Release Date	Links			
Homo sapiens (human)	GRCh38.p14 (GCF_000001405.40)	GCF_000001405.40-RS_2025_08	2025-08-01	2025-08-06	FTP	B	AR	GDV
Homo sapiens (human)	T2T-CHM13v2.0 (GCF_009914755.1)	GCF_009914755.1-RS_2025_08	2025-08-01	2025-08-06	FTP	B	AR	GDV
Danio rerio (zebrafish)	GRCz12tu (GCF_049306965.1)	GCF_049306965.1-RS_2025_04	2025-04-11	2025-05-12	FTP	B	AR	GDV
Rattus norvegicus (Norway rat)	GRCr8 (GCF_036323735.1)	GCF_036323735.1-RS_2024_02	2024-02-09	2024-02-22	FTP	B	AR	GDV
Zea mays (maize)	Zm-B73-REFERENCE-NAM-5.0 (GCF_902167145.1)	GCF_902167145.1-RS_2025_02	2025-02-14	2025-02-20	FTP	B	AR	GDV
Mus musculus (house mouse)	GRCm39 (GCF_000001635.27)	GCF_000001635.27-RS_2024_02	2024-02-01	2024-02-07	FTP	B	AR	GDV
Apis mellifera (honey bee)	Amel_HAv3.1 (GCF_003254395.2)	104	2018-09-13	2018-09-19	FTP	B	AR	GDV

Анатомия записи GenBank (Flat File)

Файл содержит не только буквы кода, но и инструкцию по их прочтению.
Раздел FEATURES — это карта.

```
LOCUS      NM_000618      333 bp  mRNA  linear  PRI 20-APR-2023
DEFINITION Homo sapiens insulin (INS), transcript variant 1, mRNA.
```

FEATURES	Location/Qualifiers
source	1..333 /organism="Homo sapiens" /mol_type="mRNA" /db_xref="taxon:9606"
gene	1..333 /gene="INS"
CDS	1..333 /gene="INS" /codon_start=1 /product="insulin preproprotein"

CDS 1..333

Координаты кодирующей
последовательности (где
зашифрован белок)

FASTA: Универсальный язык биоинформатики

Для работы алгоритмов и программ лишние метаданные не нужны.
FASTA оставляет только суть: заголовок и последовательность.

```
LOCUS       NM_000618           333 bp  mRNA  linear
DEFINITION  Homo sapiens insulin (INS), transcript
```

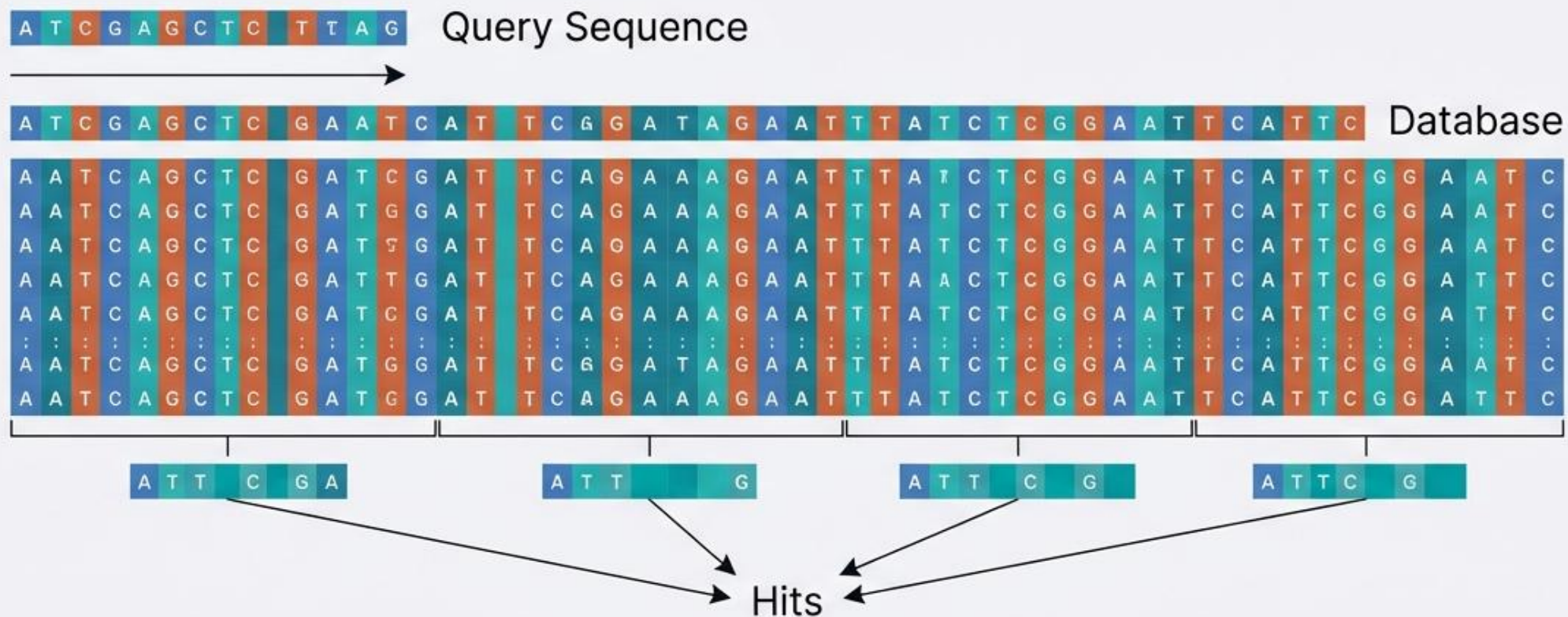
FEATURES	Location/Qualifiers
source	1..333 /organism="Homo sapiens" /mol_type="mRNA" /db_xref="taxon:9606"
gene	1..333 /gene="INS"
CDS	1..333 /gene="INS" /codon_start=1 /product="insulin preproprotein"

**Очистка от
метаданных**

```
>NM_000618.4 Homo sapiens
insulin (INS), transcript
variant 1, mRNA
AGCCCTCCAGGACAGGCTGCATCAGAA
AGAGGCCATCAAGCAGGTCTGTTCCAA
GGGCCTTTGCGTCAGGT
```

BLAST: Поиск по сходству

Алгоритм выравнивает вашу последовательность (Query) с базой данных (Database) для поиска совпадений.



Standard Nucleotide BLAST

blastn

blastp

blastx

tblastn

tblastx

BLASTN programs search nucleotide databases using a nucleotide query. [more...](#)[Reset page](#)[Bookmark](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

NM_001185098.2

Query subrange [?](#)

From

To

Or, upload file

Выберите файл Файл не выбран [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)☐ Align two or more sequences [?](#)

Choose Search Set

Database

☒ Standard databases (nr etc.): ☐ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus ☐ Experimental databases◆ Nucleotide collection (nr/nt) [?](#)

Organism

Optional

Enter organism name or id--completions will be suggested

☐

exclude

[Add Organism](#)Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude

Optional

☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to

Optional

☐ Sequences from type material

Entrez® Query

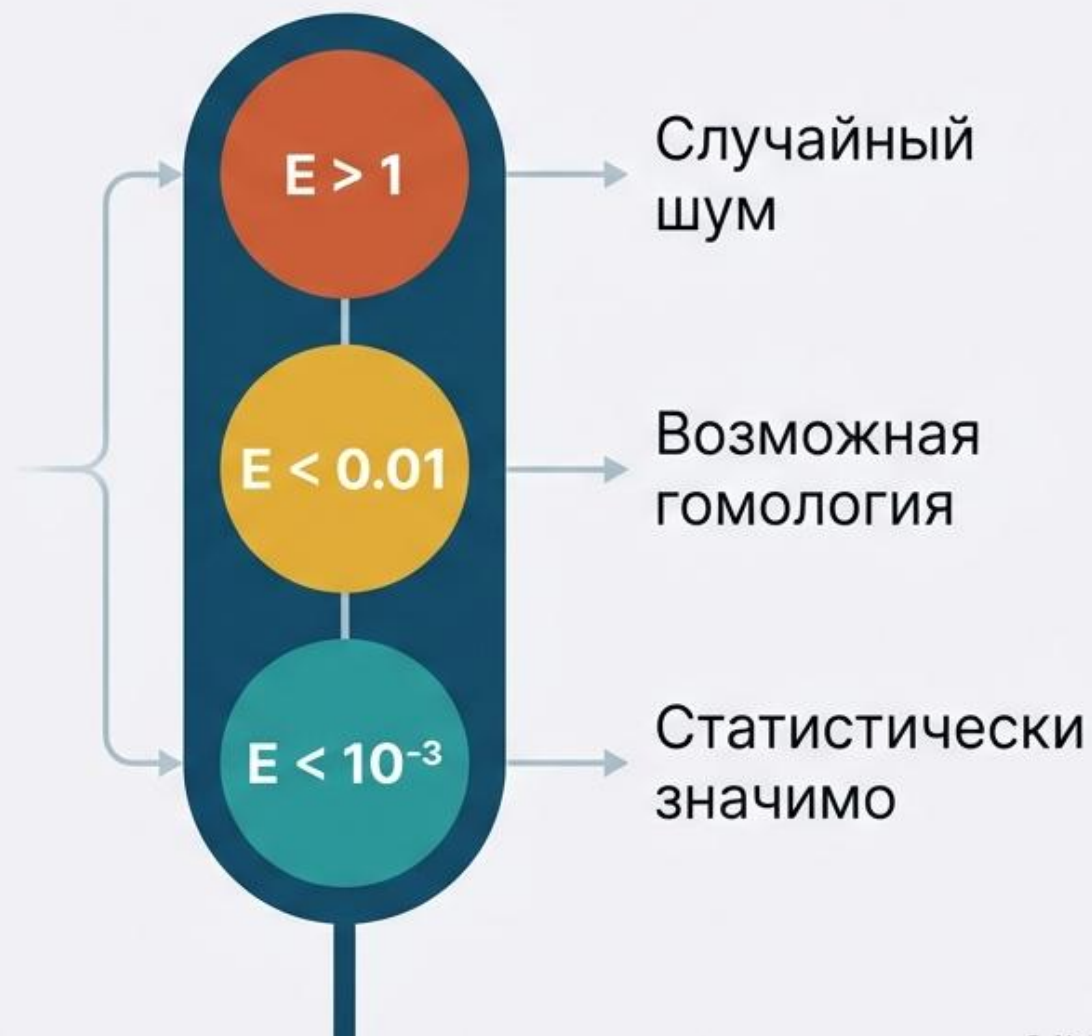
Optional

Enter an Entrez query to limit search [?](#)[YouTube](#) [Create custom database](#)

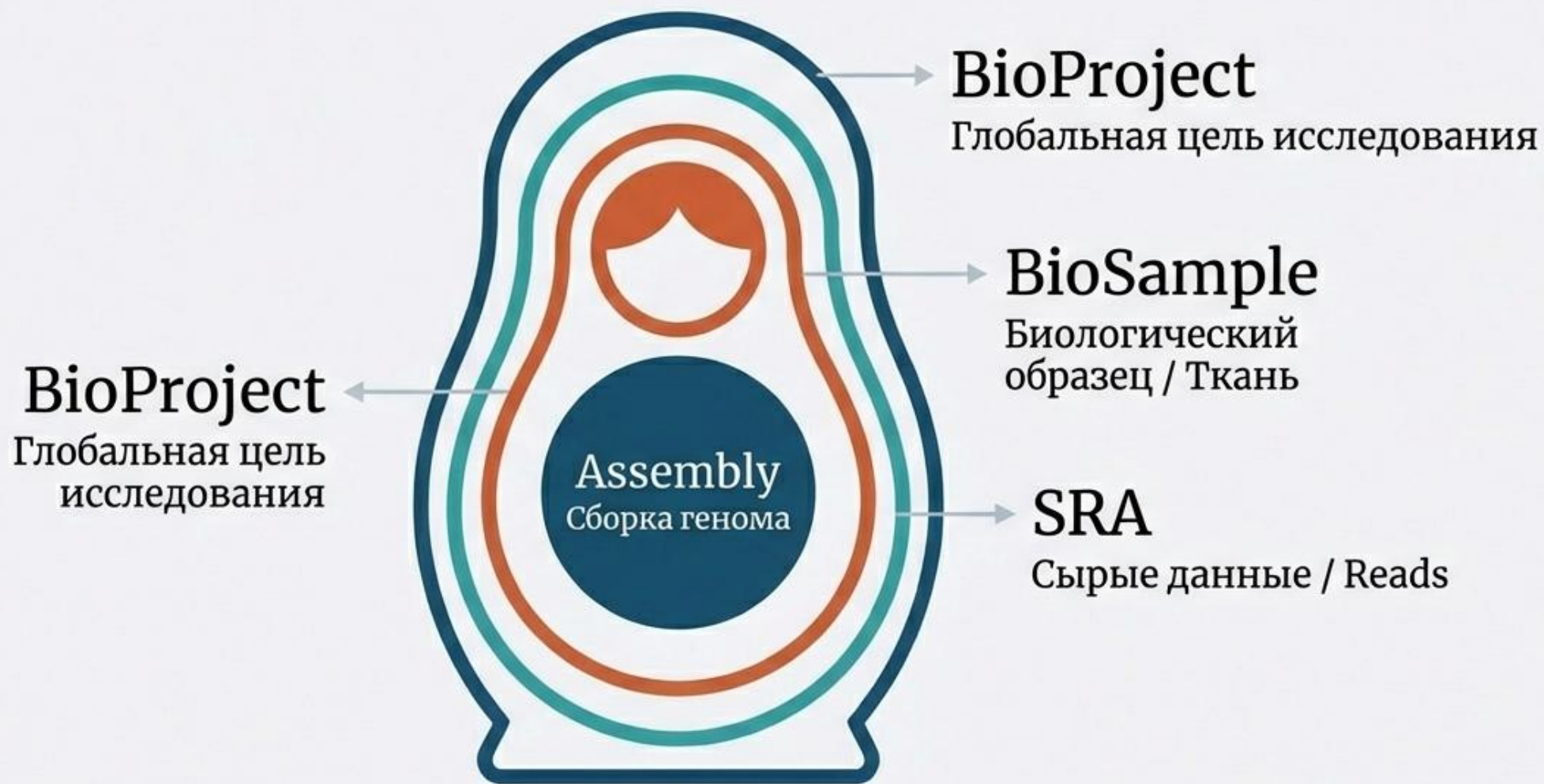
E-value: Это шум, а не вероятность

$$E = m \cdot n \cdot 2^{-s}$$

E-value — это ожидаемое количество случайных совпадений. Чем база больше, тем выше шанс случайности.

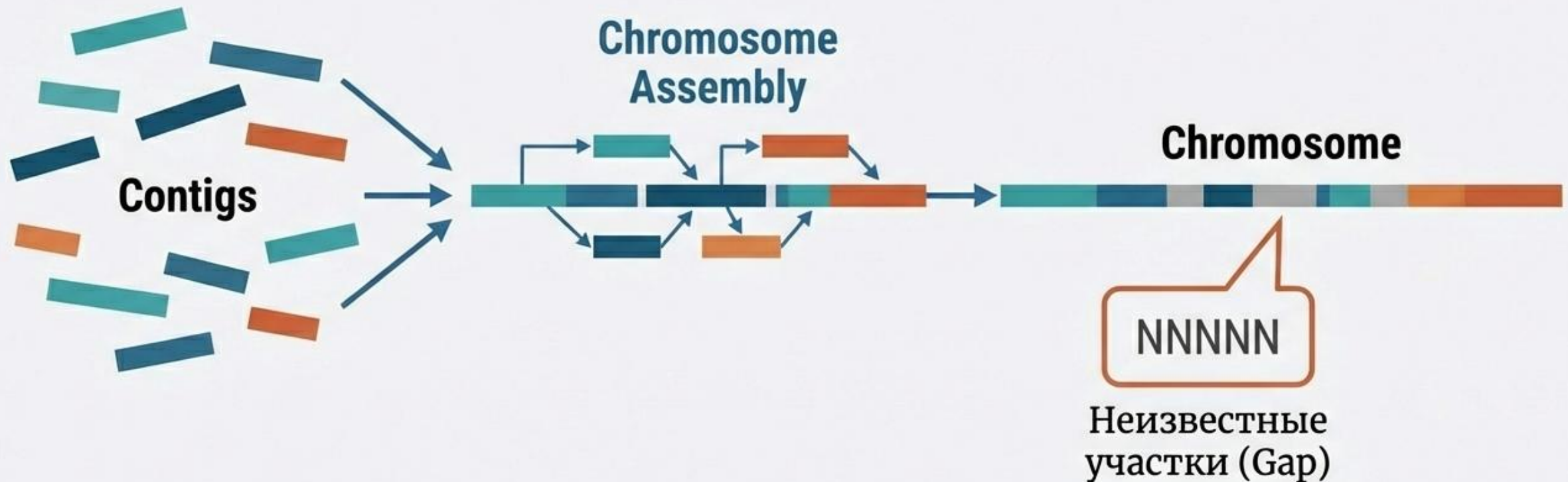


Иерархия данных: Принцип Матрешки



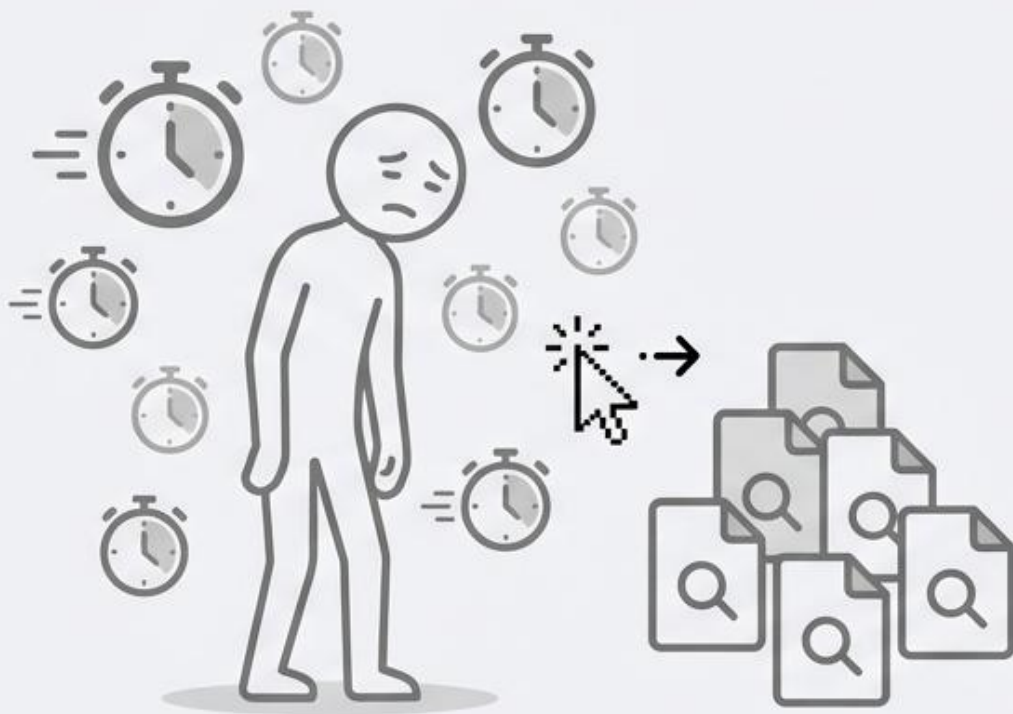
Хромосома — это виртуальный объект

Секвенаторы не читают хромосомы целиком. Запись в базе — это "склейка" (scaffolding) из множества фрагментов.



Автоматизация рутины

Когда нужно скачать 100 последовательностей, ручной поиск — это потеря времени. E-utilities (Biopython) позволяют сделать это за секунды.



Ручной поиск



Biopython

Несколько строк кода для выгрузки сотен последовательностей

Находит ID
записей

Список
идентификаторов

Скачивает
данные

Конвертирует и
сохраняет

```
from Bio import Entrez, SeqIO
```

```
Entrez.email = "ivanovii@my.msu.ru"
```

```
# Ищем ID всех записей по запросу
```

```
handle = Entrez.esearch(db="nucleotide", term="BRCA1[Gene] AND human[Orgn]",  
retmax=5)
```

```
record = Entrez.read(handle)
```

```
ids = record["IdList"]
```

```
# Качаем их разом
```

```
fetch_handle = Entrez.efetch(db="nucleotide", id=ids, rettype="fasta", retmode="text")
```

```
#Сохраняем в нужном формате
```

```
records = list(SeqIO.parse(fetch_handle, "fasta"))
```

```
SeqIO.write(records, "my_genes.fasta", "fasta")
```